

1. Institutional User Accounts & Demo Credentials

All pre-configured user accounts are initialized with standard password: **Demo@12345**

Role	Login Email	Password	Key Permissions & Access Scope
Platform Admin	admin@apnileap.org	Demo@12345	Global governance, unrestricted project creation.
Dean / Principal (KLE)	kle.dean@apnileap.org	Demo@12345	Institute governance, project creation across all departments.
HOD - CSE (AI)	kle.hod.cseai@apnileap.org	Demo@12345	CSE-AI project creation, mentor allocation, review approvals.
HOD - CSE	kle.hod.cse@apnileap.org	Demo@12345	CSE department project creation, mentor reviews, portfolio governance.
Faculty Mentor (CSE-AI)	sujata.kotabagi@kletech.ac.in	Demo@12345	Assigned CSE-AI projects, milestone & KPI updates, challenge logging.
Faculty Mentor (CSE)	amit.kachavimath@kletech.ac.in	Demo@12345	Assigned CSE projects, milestone & KPI updates, challenge resolution.
Student Account (Demo)	01fe23bcs001@kletech.ac.in	Demo@12345	Project workspace view, task movement, evidence submission.

2. Step-by-Step Instructions to Insert a New Project

- **Step 1: Sign in with a Governance Account:** Log in at <http://localhost:4000> with kle.hod.cseai@apnileap.org or kle.dean@apnileap.org (Password: Demo@12345).

• **Step 2: Navigate to Target Department:** Click on the Department, select the Theme, then click '+ Add Project'.

• **Step 3: Enter Project & Team Details:** Copy and paste Title, Artefact, Mentor Name, Academic Year (2026-27), Semester (Sem-5), and 4 student SRNs from below.

• **Step 4: Save & Automated Jira Provisioning:** Click 'Save Project'. ApniLeap provisions the Jira Cloud board, seeds tasks, and creates Jira-linked KPIs.

• **Step 5: Verify on Dashboard & Workspace:** Click the '[Jira Board]' button to open the live board, or open the workspace to interact with tasks.

3. Project 1: Autonomous Drone Edge-AI Pipeline for Crop & Canopy Health

Form Field	Value to Copy & Paste
Department	Computer Science and Engineering (AI)
Theme	Edge AI & Computer Vision
Project Title	Autonomous Drone Edge-AI Pipeline for Precision Crop & Canopy Health Monitoring
Artefact Title	Edge-Deployed Aerial Multispectral Crop Health Inspector
Academic Year / Sem	2026-27 / Sem-5
Faculty Mentor	Dr. Sujata Kotabagi
Student 1 (Lead)	Rohan Deshpande SRN: 01FE23BCS961 Div: A Sem: Sem-5
Student 2	Ananya Kulkarni SRN: 01FE23BCS962 Div: A Sem: Sem-5
Student 3	Varun Hegde SRN: 01FE23BCS963 Div: A Sem: Sem-5
Student 4	Kavya Patil SRN: 01FE23BCS964 Div: A Sem: Sem-5

Project 1 Challenges to Log:

Challenge Title	Root Cause & Impact	Action Plan & Owner
Drone Edge Compute Thermal Throttling	Kernel load causes Jetson Orin to hit 85°C. Video drops to 14 FPS, skipping canopy sections.	Cap batch frequency at 35 FPS, apply heat-pipe cooling. Owner: Rohan Deshpande

NDVI Calibration Drift Under Variable Sun	Cloud variability changes spectral intensity. NDVI deviates by 22% triggering false alerts.	Normalize reflectance spectra frame-by-frame using downwelling sensor. Owner: Ananya Kulkarni
Telemetry Packet Loss Over 2.4GHz Link	Dense tree canopy attenuates 2.4GHz signal, losing telemetry heartbeat at boundaries.	Switch link to 868MHz high-gain antenna with packet retransmission. Owner: Varun Hegde

Project 1 KPIs to Enter:

KPI Name	Target	Measured	Evidence Note
Canopy Anomaly Detection Latency	35 ms	31.4 ms	TensorRT benchmark output logged across 1,000 flight images.
NDVI Vegetation Classification Accuracy	93.5 %	94.8 %	Validated against ground-truth agronomist inspection samples.
Aerial Video Ingestion Stream Rate	45 fps	46.2 fps	DeepStream pipeline telemetry logs during 20-minute flight.
Drone Embedded Compute Power Draw	14 Watts	12.6 Watts	Power rail telemetry recorded via Jetson jtop utility.

4. Project 2: Decentralized Campus Microgrid Energy Distribution Engine

Form Field	Value to Copy & Paste
Department	Computer Science and Engineering
Theme	Sustainable Campus & Smart Energy
Project Title	Decentralized Campus Microgrid Energy Distribution & Load Forecasting Engine
Artefact Title	Smart Microgrid Energy Balancer & Predictive Dispatcher
Academic Year / Sem	2026-27 / Sem-5
Faculty Mentor	Amit Kachavimath
Student 1 (Lead)	Nikhil Rao SRN: 01FE23BCS971 Div: B Sem: Sem-5
Student 2	Deepa Shrestha SRN: 01FE23BCS972 Div: B Sem: Sem-5
Student 3	Karthik Bhat SRN: 01FE23BCS973 Div: B Sem: Sem-5
Student 4	Meera Kulkarni SRN: 01FE23BCS974 Div: B Sem: Sem-5

Project 2 Challenges to Log:

Challenge Title	Root Cause & Impact	Action Plan & Owner
Solar Inverter Telemetry Desynchronization	RS485 polling cycle lags during rapid solar insolation drops, causing substation power factor dips.	Migrate from 1000ms serial to 50ms event-driven Modbus-TCP socket. Owner: Nikhil Rao
Peak-Hour Load Forecasting Error > 10%	ARIMA model fails to capture chiller compressor start-stop cycles, discharging batteries sub-optimally.	Train Temporal Fusion Transformer model incorporating timetable features. Owner: Deepa Shrestha
MQTT Telemetry Dropping Under Wi-Fi Traffic	Substation smart meters share 2.4GHz Wi-Fi with campus mobile traffic during peak lunch breaks.	Enable MQTT QoS Level 1 with persistent local buffer queue. Owner: Karthik Bhat

Project 2 KPIs to Enter:

KPI Name	Target	Measured	Evidence Note
Microgrid Short-Term Load Forecast MAPE	4.5 %	3.8 %	Scikit-learn evaluation report comparing predicted vs actual readings.
Automated Phase Balancing Response Time	250 ms	185 ms	Inverter timestamped telemetry log recording transient load step responses.
Solar Inverter Ingestion Reliability	99.8 %	99.92 %	PostgreSQL time-series database continuous ingestion audit query for 7 days.

Peak Load Demand Reduction	18.0 %	19.4 %	Substation billing report comparing peak demand charges before/after dispatch.
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5. Post-Creation Verification Checklist

- [✓] **Jira Cloud Board:** Click [■ Jira Board ■] in the project header to confirm the dedicated software board opens.
- [✓] **Workspace Kanban Tasks:** Verify 3 tasks appear with active transition buttons (← To Do, In Progress →, Completed ✓) and Jira issue links.
- [✓] **KPI Hyperlinks:** Open the KPIs tab to verify that starter and newly added KPIs display clickable ■ KEY-N ■ badges.
- [✓] **Challenge Hyperlinks:** Log any sample challenge and confirm an associated Jira issue is created with a live link.